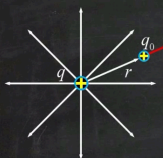


Title:

Date: / /

考点	重要程度	占 分	常见题型
1. 离散型	★★	0-3	选择/填空
2. 连续型	★★★★★	5-10	选择/填空/大题
3. 结论型			

1. 求电场强度—离散型



$$F = \frac{1}{4\pi\epsilon_0} \frac{q \cdot q_0}{r^2}$$

$$E = \frac{F}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\text{库伦定律: } F = \frac{1}{4\pi\epsilon_0} \frac{q \cdot q_0}{r^2}$$

- (1) 同性相斥，异性相吸
- (2) 电场力跟 q, q_0 电量成正比，跟 r^2 成反比
- (3) 真空介电常量: $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N} \cdot \text{m}^2)$

$$\text{电场强度: } E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \quad (\text{矢量, 有大小有方向})$$

- (1) $\vec{E}$ 方向判定: 将正电放入电场中, 受力方向即 $\vec{E}$ 的方向.
- (2) 电场强度跟场源电荷 q 和位置 r 有关, 与试探电荷 q_0 无关.

2. 求电场强度—连续型

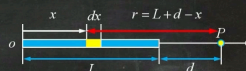
$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \quad E = \int dE = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

题1. 均匀带电细棒, 棒长 L , 电荷线密度 λ , 求: 棒的延长线上与棒的右端相距 d 处的场强.

解: 在带电棒上取一线元 dx

$$\text{则 } dx \text{ 所带的电荷量: } dq = \lambda \cdot dx$$

$$\text{则在 } P \text{ 处: } dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda \cdot dx}{(L+d-x)^2}$$



$$E = \int dE = \int_0^L \frac{1}{4\pi\epsilon_0} \frac{\lambda}{(L+d-x)^2} dx$$

$$= \frac{\lambda}{4\pi\epsilon_0} \left[\frac{1}{L+d-x} \right]_0^L = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{L+d} - \frac{1}{L} \right)$$

2. 求电场强度—连续型

$$dE = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \quad E = \int dE = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2}$$

题2. 一个半径为 R 的均匀带正电半圆环, 电荷密度为 λ , 求环心处 O 点的场强和方向

解: 在半圆环上任取线元 dl

$$dl \text{ 所带的电量: } dq = \lambda \cdot dl$$

$$\text{在 } O \text{ 点产生的场强: } dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda \cdot dl}{R^2} = \frac{\lambda}{4\pi\epsilon_0 R^2} dl$$

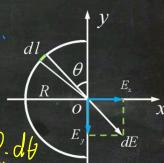
$$dE_x = dE \cdot \sin \theta = \frac{\lambda}{4\pi\epsilon_0 R^2} \sin \theta dl$$

$$dE_y = dE \cdot \cos \theta = \frac{\lambda}{4\pi\epsilon_0 R^2} \cos \theta dl$$

$$E_x = \int dE_x = \int \frac{\lambda}{4\pi\epsilon_0 R^2} \sin \theta dl = \frac{\lambda}{4\pi\epsilon_0 R^2} \int \sin \theta \cdot R d\theta = \frac{\lambda}{2\pi\epsilon_0 R}$$

$$E_y = \int dE_y = \int \frac{\lambda}{4\pi\epsilon_0 R^2} \cos \theta dl = \frac{\lambda}{4\pi\epsilon_0 R^2} \int \cos \theta \cdot R d\theta = 0$$

$$E = E_x = \frac{\lambda}{2\pi\epsilon_0 R} \quad \text{方向朝 } x \text{ 正方向}$$



3. 求电场强度—结论型

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 x} \sin \theta_1$$

$$E = \frac{\lambda}{2\pi\epsilon_0 R} \sin \theta_1$$

$$E = \frac{q\lambda}{4\pi\epsilon_0 (R^2 + x^2)^{3/2}}$$

$$E = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right]$$

$$E = \frac{\sigma}{2\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 x}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad (r > R)$$

$$E = 0 \quad (r < R)$$

$$E = \frac{q}{4\pi\epsilon_0 r^2} \quad (r > R)$$

3. 求电场强度—结论型

题3. 一宽为 b 的无限长均匀带电平面薄板, 其电荷面密度为 σ , 如图所示. 试求平板所在平面内, 距薄板边缘为 a 处的电场强度.

解: 薄板内任取一条, 宽为 dx

$$\text{则窄条可视为无限长带电电线}$$

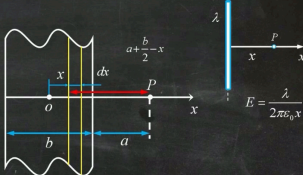
$$\text{线密度: } \lambda = \sigma dx$$

无限长带电电线在 P 点的场强:

$$dE = \frac{\sigma dx}{2\pi\epsilon_0 (a + \frac{b}{2} - x)}$$

整个薄板在 P 点的场强:

$$E = \int dE = \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{\sigma dx}{2\pi\epsilon_0 (a + \frac{b}{2} - x)} = \frac{\sigma}{2\pi\epsilon_0} \ln \frac{a+b}{a}$$



蜂考

蜂考

蜂考

蜂考

蜂考

蜂考

$$d\vec{E} = \frac{dq}{4\pi\epsilon_0 r^2}$$

$$\vec{E} = \int d\vec{E} = \int \frac{dq}{4\pi\epsilon_0 r^2}$$

$$\vec{E} = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right]$$

$$\frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{R^2 + x^2}} \right]$$

$$E = \frac{q\lambda}{4\pi\epsilon_0 (R^2 + x^2)^{3/2}}$$

$$\vec{E} = \frac{\sigma}{2\epsilon_0}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 x}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\vec{E} = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\vec{E} = \begin{cases} 0 & (r < R) \\ \frac{q}{4\pi\epsilon_0 r^2} & (r > R) \end{cases}$$

$$\vec{E} = \begin{cases} \frac{q\lambda}{4\pi\epsilon_0 R^2} & (r < R) \\ \frac{q}{4\pi\epsilon_0 r^2} & (r > R) \end{cases}$$

$$E = \int \frac{\lambda r}{2\pi\epsilon_0 R^2} \quad (r < R)$$

$$\frac{\lambda}{2\pi\epsilon_0 r} \quad (r > R)$$